Universality in Quantum Mechanics:
A Claim That Cannot Be Tested, and One That Can

Claes Johnson[1]

[1]KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden ,
claesjohnson@gmail.com

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Abstract

Standard quantum mechanics is held to be universal for chemistry: one parameter-free equation, with the nuclear charges as its only input, determining every atom, molecule and material. We ask on what grounds that claim rests and find that it cannot be tested. The equation is a problem in $3N$ dimensions whose cost grows exponentially in the electron number, so beyond a few tens of electrons it is uncomputable, and what is computed in its place is not the equation but a reduction carrying parameters: basis sets, truncation orders, exchange–correlation functionals. Agreement between a parameterised reduction and observation does not transfer to the parameter-free equation behind it, and the hierarchy of approximations cannot carry it upward, since the hierarchy's exact limit is precisely the uncomputable object while every reachable rung is parameterised. The universality of standard quantum mechanics is therefore verified at one to three electrons and assumed everywhere else, with the assumption shielded from refutation by the intractability of the thing assumed. We then ask the same question of RealQM, a model of non-overlapping unit electron charge densities in three space dimensions, equally parameter-free with the nuclear charge as its only input. Because RealQM is computable at the scale of real systems — cost linear in the electron number — its universality is an empirical claim, and we report the record: atoms from hydrogen to radon, the row capacities of the periodic table, the H_2 bond at 94% of the measured dissociation energy, water's dipole and bond angle to 99% and 97%, hydrogen-bond geometries in dimers and ice matching measurement, metallic bonding, reaction paths through transition states, and protein folds to 153 residues on a laptop. The comparison that matters is the last: the method held to be universal cannot fold a protein at all. The asymmetry is dimensional rather than technical — a model in three dimensions can be held to account over the scope it claims, a model in $3N$ cannot.

1 Two parameter-free models

Two models of electronic matter are on the table, and both are parameter-free in the same sense: the nuclear charge Z is supplied and nothing is fitted.

Standard quantum mechanics. The N-electron Schrödinger equation. The state is a complex antisymmetric function $\Psi(x_1, \ldots, x_N)$ on $\mathbb{R}^{3N}$, the Hamiltonian is kinetic energy plus Coulomb interaction among electrons and nuclei, and no constant appears that was not already in the description of the charges.

RealQM. The state is a collection of N one-electron charge densities on non-overlapping domains in $\mathbb{R}^3$, continuous across free interfaces, interacting by Coulomb potentials without self-repulsion, the ground state obtained by minimising the energy. Again nothing beyond Z.

The models differ in the dimension of the object they carry: $3N$ against 3. The burden of this article is that this is not a difference of convenience.

2 What a universality claim asserts

A model is universal over a domain of phenomena if three conditions hold.

1. **Species-blindness.** The model makes no reference to the kind of system it is applied to. It does not mention atoms, bonds, metals or proteins; one writes the same thing for helium and for a protein, differing only in Z and the geometry.

2. **Closure under composition.** Joining two systems yields the same model for the union. There is no boundary in the domain at which one must change theories.

3. **Accountability.** The model's predictions can be compared with measurement across the domain claimed, and the comparison can go against it.

The first two are formal and easily checked; both models satisfy them completely. The third is not formal at all, and it is the one that does the work. A model satisfying only the first two is a *framework*, not a universal theory, because nothing it says about most of its domain can be held to anything.

3 Why the universality of standard quantum mechanics cannot be tested

3.1 The equation is uncomputable where the claim is interesting

The state is a function on $\mathbb{R}^{3N}$, and representing it costs exponentially in N: full configuration interaction in a basis of M functions scales as $\binom{M}{N}^2$ in the worst case, and the exact solution is in practice out of reach past roughly twenty electrons. This is not a matter of waiting for better machines. The ground-state problem for a local Hamiltonian is QMA-complete, so no efficient general algorithm is believed to exist even on a quantum computer; the barrier has the character of a theorem rather than of an engineering limit.

The systems for which universality is an interesting claim — a catalyst, a protein, a crystal, a cell — carry 10^3 to 10^{23} electrons. For every one of them the model cannot be solved.

3.2 What is computed instead is not the model

Since the equation cannot be solved, it is not the equation that is compared with experiment. What is compared is a reduction, and every reduction in use carries parameters. Gaussian basis sets are built by optimising exponents and contraction coefficients atom by atom, and the complete-basis-set limit is reached by extrapolation through a fitted functional form. Coupled-cluster theory truncates the excitation operator, and CCSD(T)'s perturbative triples correction is not variational and carries no bound — its celebrated kcal/mol accuracy is an empirical regularity over well-behaved closed-shell molecules which degrades without warning for multireference

systems. Density functional theory, which performs essentially all practical calculation on large systems, selects an exchange–correlation functional from a zoo in which B3LYP, PBE, M06 and ωB97X are different theories with different failures, several fitted to benchmark sets outright.

We should be fair about the strongest case. There is a window — small closed-shell molecules of up to a few tens of atoms — in which CCSD(T) near the basis-set limit is nearly parameter-free and agrees with measurement to about a kcal/mol across a wide range of chemistry. That is a real achievement and we do not dispute it. But it is a window, and outside it, which is to say for almost everything, the computed object is a functional-bearing reduction.

3.3 The hierarchy cannot transfer verification upward

It is often said that this does not matter, because the approximations form a convergent hierarchy — Hartree–Fock, MP2, CCSD, CCSD(T), full CI — whose limit is the exact solution, so the hierarchy connects what is computed to what is claimed.

It does not, for a structural reason. The hierarchy has two ends and both are unavailable. Its exact limit is the uncomputable object itself, so one can never stand at the top and check. Its lower rungs, which are what gets computed, each carry parameters, so agreement there is agreement of a parameterised procedure. There is no rung which is both reachable and parameter-free, and consequently no step at which empirical success attaches to the parameter-free equation rather than to the reduction. The existence of a convergent sequence establishes that the reductions approach something; it does not establish that the something is nature, since that identification is exactly what was to be tested.

3.4 The practice this produces

The consequence is visible in how discrepancies are handled. When density functional theory fails on dispersion — and semilocal and hybrid functionals do fail on it, systematically — the response is to add an empirical correction, a fitted term, and the failure is charged to the functional. When a coupled-cluster result is wrong for a transition-metal complex it is charged to the truncation or the basis. In no case is it charged to the Schrödinger equation, and this is not dogmatism: it is forced, because no computation of the equation exists against which the equation could be found wanting.

A claim that no measurement can contradict is not thereby strong; it has been removed from the class of claims that measurement bears on. Over the great majority of its asserted domain that is the position of the universality of standard quantum mechanics — verified superbly at one to three electrons, assumed beyond.

4 Why the universality of RealQM can be tested

RealQM is in the opposite position on the one point that matters, for one reason: it is computable where the claim is interesting. The state is N densities on $\mathbb{R}^3$, so the cost is the electron number times the number of mesh points — linear in N, not exponential. A molecule, a cluster, a crystal, a protein is the same computation with more kernels and more domains. The model itself, with Z as its only input and no reduction interposed, can be run on the systems in question and the output compared with measurement.

That makes universality an empirical question in the sense of Section 2(3). What follows is the record. It is organised by scale, because the point is not any single number but that the

same four-line scheme, unchanged, reaches every scale at which chemistry happens.

4.1 Atoms

Hydrogen is an identity, not a test. For one electron there is no exclusion and no partition, and RealQM *is* the Schrödinger equation: both give $E = -0.5$ Ha and $\alpha = 4.5\, a_0^3$. We record it because it locates where the two models can first differ — at two electrons, where the exclusion postulate enters.

Helium. The first real test, with the answer known exactly.

	E (Ha)	error
exact (nonrelativistic)	-2.9037	—
RealQM, two domains, front-tracked plane	-2.91	0.24%
Hartree–Fock	-2.8617	1.4%

The two-domain result is closer to exact than Hartree–Fock, recovering about two thirds of the correlation energy without any correlated wave function. Territorial exclusion, replacing antisymmetry, is not merely tolerable on helium's energy; it is better than the mean-field theory it stands beside. Lithium and the lithium cation follow the same pattern against their exact values of -7.478 and -7.280 Ha.

The table. Computed all-electron on the bare kernel $-Z/r$, with no core radius, no fitted length and nothing adjusted per element: through argon the mean relative error is 1.6% with maximum 4.5%, and 13 of 17 atoms fall within 2.5%. Extended to the fourth row on a mesh chosen by Z, the mean is 3.3%, every entry under-bound in the direction the spherical reduction predicts. Energies are available through to radon.

The periodic table's capacities. Two filled packings appear and no third: the antipodal pair, 2, and the octet, $8 = 4{+}4$, two dual tetrahedra. A lone tetrahedron has no inversion centre and so cannot close a shell; its dual completes the cube, which is centrosymmetric and admits no further dual. Row capacities are sums of these two alone,

$$2, \qquad 8, \qquad 18 = 2{+}8{+}8, \qquad 32 = 8{+}8{+}8{+}8,$$

and a period ends when its capacity is exhausted, giving the closures $2, 10, 18, 36, 54, 86, 118$. The capacities obey a parity rule — remainder modulo 8 is only ever 0 or 2, never 4 or 6, so each is a whole number of octets carrying at most one pair, the pair appearing exactly when the shell index is odd — and admit a closed form, $\mathrm{cap}(n) = 8\lfloor n/2 \rfloor \lceil n/2 \rceil + 2(n \bmod 2)$, which equals $2n^2$ for every n without ever invoking it. The account coincides with the hydrogenic degeneracy rather than borrowing it.

Spectra and valence quantities. Excited-state spectra are computed for the alkalis, the alkaline earths and helium, including the helium ground-state-to-orthohelium excitation, and ionization energies for lithium and fluorine with their ions. Electronegativity comes out natively: Allen's configuration energy reproduces the ordering across period 2 exactly.

4.2 Molecules

The covalent bond. H_2 on a 200^3 grid, with the binding energy taken self-referentially from the same solver as $E_{\text{bind}} = E(R_e) - E(R{=}6)$:

quantity	RealQM	reference
E at $R = 2.0\,a_0$	-1.1923 Ha	-1.1745 Ha (Kolos–Wolniewicz)
E at $R = 6\,a_0$	-1.0379 Ha	-1.000 Ha ($2\times$H)
E_{bind}	-0.1543 Ha (-4.20 eV)	-0.1646 Ha (measured D_0)

94% of the measured dissociation energy, from a model with no basis set and no antisymmetry. The curve dissociates correctly to two hydrogen atoms as R is increased, so the bond is formed and broken by the same scheme. Worth noting on what is being compared: of the H_2 curve only R_e and D_0 are measured quantities — the energy at arbitrary R is itself computed. The textbook six-decimal agreement for H_2 belongs to a computation, not to a measurement.

Water. The dipole moment and the bond angle are the two quantities that test whether the charge is in the right place. With the oxygen valence split into angular sectors, one toward each hydrogen and one pointing away:

quantity	RealQM	experiment	ratio
dipole moment μ	1.836 D	1.85 D	99%
H–O–H angle	101.6°	104.5°	97%
equilibrium O–H bond	$1.814\,a_0$	$1.814\,a_0$	exact

The lone pair is not put in by hand; the sector pointing away from both hydrogens is where the energy puts it. The equilibrium bond length comes out on the measured value.

4.3 Condensed matter and reactions

The hydrogen bond is where molecular matter becomes material, and it is geometric:

system	RealQM	reference
water dimer	H$\cdots$O = 1.87, O–O = 2.95 Å	matches measurement
formamide dimer	N–H = 1.01, H$\cdots$O = 2.04, N$\cdots$O = 2.97 Å	matches measurement
ice I_c, 216 molecules	O–O = 2.73 Å	2.76 Å

Beyond hydrogen bonding, the same scheme gives metallic bonding — a 100-atom lithium lattice, perturbed, restores itself — ionic crystals and their melting in sodium chloride, conduction barriers for distinct arrangements of charge, and chemical reaction paths: H + H_2 → H_2+ H proceeds by bond exchange through the H_3 transition state, with proton transfer, ion formation and salt dissolution computed on the same footing.

4.4 Biomolecules

Nothing in the model changes at this scale; only the number of domains.

system	result	reference
DNA G:C base pair	2 of 3 Watson–Crick bonds at ~ 2.1 Å	1.9 Å
RNA hairpin, 12 nt	all 4 stem pairs form, G1:C12 = 1.99 Å	2.0 Å
Trp-cage, 20 residues	4 of 5 helix H-bonds, $R_g = 7.2$ Å	near native
Villin HP35, 35 residues	3 helices, H-bonds at 2.0 Å, core F10–F17 = 6.1 Å	3-helix bundle
Myoglobin, 153 residues	helices A = 1.51, G = 2.15, H = 2.68 Å	partial fold
coiled coil, 2×15 residues	helix separation $12 \rightarrow 7.5$ Å, blind	water-driven docking

The RNA hairpin assembles its own secondary structure from a U-shaped start: stem paired, loop open, correct topology. The coiled-coil docking is blind — no bias toward the bound state — and is driven by the water.

4.5 The scope is exercised, not asserted

This is the comparison that bears directly on universality, because it is about what each method can be *held to*:

method	largest fold	hardware	time
Gaussian / VASP (DFT, HF)	**cannot fold proteins**	HPC cluster	days per point
GROMACS / OpenMM (MD)	20 residues	GPU clusters	weeks to months
Anton 2 (MD)	80 residues	$100M custom ASIC	months per protein
RealQM	**153 residues**	laptop GPU	minutes to hours

The quantum-chemical methods — the ones carrying the universality claim — appear in the first row, and they do not reach the problem. The two middle rows reach it with empirical force fields fitted to experiment and quantum calculation, so they are not tests of quantum mechanics at all. RealQM reaches it from the charges and Z, on commodity hardware. Whatever one concludes about its accuracy, its scope is exercised rather than asserted, and that is the condition Section 2(3) asks for.

5 Scope and fidelity

One bound must be stated, because it marks the limit of what the record above establishes. RealQM is a fixed ansatz. Refining its mesh converges to the ansatz's own answer, not toward nature's, and there is no next order to add — no analogue of going from CCSD to CCSD(T) to full CI. So the claim supported here is universality of **scope**: the model applies across every scale of chemistry, is checked there, and agrees to a few per cent on energies and considerably better on geometry. It is not a claim of universal **fidelity**, which would require a convergence ladder that the model does not have. Being a testable claim, it is also refutable, and in places it has been refuted; that is the price and the point of being accountable.

6 The asymmetry is dimensional

It would be easy to read the foregoing as a comparison of accuracy, and that is not the point. The two universality claims are of different *kinds*, and the dimension of the state is what makes them so.

A model whose state lives in $3N$ dimensions cannot be evaluated on a system of chemical interest, so its universality is a statement about an object that is never exhibited — an existence claim without a witness. A model whose state lives in 3 dimensions can be evaluated, so its universality is a prediction, and predictions can fail. This is the same distinction that separates a non-constructive existence proof from a computed solution, and it recurs across applied mathematics: the symbolic object is more general and cannot be inspected, the computed object is specific and can be checked. Neither is worthless, but they are not interchangeable, and universality means something different in each.

	standard QM	RealQM
state	$3N$ dimensions	3 dimensions
parameters	none, in the equation	none
cost	exponential in N	linear in N
species-blind	yes	yes
closed under composition	yes	yes
reaches a protein	no	yes, 153 residues
verified range	1–3 electrons	H to Rn, and to 153 residues
verified precision	few parts in 10^5	few per cent; geometry better
exact in principle	yes	no
convergence ladder	yes, but unreachable	none
universality	**assumed**	**tested**

Standard quantum mechanics is verified at three electrons and asserted at 10^{23}. RealQM is verified from hydrogen to a 153-residue protein. Over the range where either is actually used, the second is the better evidenced — a remark about evidence, not about truth.

7 What this establishes

It does not establish that standard quantum mechanics is false. Where it can be computed it is superb, and nothing here gives any reason to doubt it.

What it establishes is narrower and more useful. The conventional ranking treats the $3N$ formulation as the serious theory and a three-dimensional model as an approximation to be tolerated. On the specific question of universality that ranking inverts, because universality is a claim about scope and accountability, and only one of the two models is accountable over the scope it claims. A framework that applies in principle to everything and can be checked against nothing has a weaker title to universality than a model which applies to everything, is checked from the helium atom to a folded protein, and agrees.